

Cover letter — *Review of Economic Studies*

Editor
Review of Economic Studies

Dear Editor,

We are pleased to submit “Volcanic Forcing and the Pre-Industrial Malthusian Trap: Pathway-Heterogeneous Demographic Response and the Long Shadow of Seasonality” for consideration at the *Review of Economic Studies*.

The paper assembles a new monthly paleo-economic panel covering 196 countries and 3,187 sub-national units back to 1421 CE—combining the ModE-RA paleo-reanalysis, HYDE 3.5 land-use reconstruction, eVolv2k volcanic sulfur record, Brecke conflict catalogue, and Allen-Nuffield wage archive—and uses volcanic eruptions as an exogenous source of climate forcing to identify the pathway-heterogeneous Malthusian response of pre-industrial economies. The paper makes three substantive contributions and one methodological one.

First, a three-equation joint panel VAR on the 1500–1900 sample identifies a sharply pathway-heterogeneous demographic margin. Crop-dominant late and pastoral/mixed late countries lose population at -9.7 and $-5.5 \cdot 10^{-5}$ per Tg of stratospheric sulfur injection respectively, both Bonferroni-significant across twelve pathway-by-equation tests, both robust to re-clustering the pathway typology on climate primitives orthogonal to HYDE outcomes, and both surviving a Webb wild-cluster bootstrap. An annual European mortality-and-fertility panel 1751–1900 attributes the demographic response to the *positive* check on infant and child mortality (the historically documented summer-mortality pattern), with the channel collapsing to zero by 1950–2022 as sanitation develops. The Allen real-wage panel returns the canonical Malthusian price-on-population coefficient at city resolution, the welfare-side echo of the demographic finding.

Second, pre-industrial inter-annual climate volatility 1421–1750 is the single strongest cross-country predictor of modern population growth at $R^2 = 0.45$, surviving both an extended deep-determinants battery and a latitude control, and failing a placebo when computed over the modern 1950–2008 reanalysis window. Seasonality, properly measured, casts a long shadow on the modern demographic trajectory that is not contained in modern climate or geographic deep determinants.

Third, we test for a Boserupian intensification channel on cropland and report an honest two-axis null on HYDE: the cropland-share specification is artifactually positive (driven by colonial-transition denominator instability in 24 small-island countries; the coefficient vanishes on the substantively-named 21-country Mediterranean / Eurasian-steppe / SE-Asia core), and the cropland-area specification is fully mediated by the demographic margin because HYDE’s reconstruction takes population as an input. Cross-validating against the independent KK10 reconstruction (Kaplan et al., 2011)—methodologically population-independent—surfaces a small but statistically robust positive signal that survives the demographic-margin control: $+1.07 \cdot 10^{-5}$ per Tg on the 21-country core ($p = 0.005$), $+4.3 \cdot 10^{-5}$ on the high-density intensive pathway ($p = 0.047$). The Boserupian

channel exists, but at a magnitude two orders below what the HYDE artefact had originally suggested, and only identifiable on a population-independent reconstruction.

The methodological contribution is the documentation of HYDE's two-axis identification problem. HYDE-based volcanic-forcing regressions appear elsewhere in the climate-economics literature without these diagnostics; our share-decomposition exercise and the HYDE-vs-KK10 cross-validation surface a measurement issue that is structurally important for the field. Combined with the demographic-margin headline (which does *not* suffer from this problem) and the long-shadow result, the paper is in a defensible position for both substantive and methodological referee evaluation.

We see the paper as a contribution to three lines of work the *Review of Economic Studies* has shaped: the macroeconomic consequences of climate and climate-volatility shocks; the structural transition from stagnation to growth, including the long shadow of pre-industrial endowments on modern outcomes; and the data-infrastructure foundations of pre-industrial comparative economics. We argue the within-country sub-national identification and the cross-reconstruction validation strategy together demonstrate the level of econometric and measurement rigor a top-5 referee round requires.

The paper is solely-authored work by two researchers at the Centro de Investigación Económica, ITAM, and Universidade de Vigo / RGEA. We declare no conflicts of interest. The manuscript has not been previously published and is not under consideration elsewhere. A replication package with the ModE-RA aggregation code, the KK10 country aggregation script, the country and sub-national parquet panels, and all analysis scripts will be deposited on OSF upon acceptance.

Sincerely,

Jorge Alonso Ortiz

Centro de Investigación Económica, ITAM | jorge.alonso@itam.mx

José-María Da-Rocha

Universidade de Vigo & RGEA, Spain | darocha@uvigo.es

Suggested referees (independent of the authors):

T. B. Andersen (Copenhagen, climate-growth); A. Matranga (NES, seasonality-agriculture); Q. Ashraf (Williams, long-run development); M. Burke (Stanford, climate-economy); J. O. Kaplan (HKUST, KK10 land-use reconstruction).